PARTIAL DIFFERENTIAL EQUATIONS

Real world problems in general involve functions of several (independent) variables giving rise to partial differential eq^{n}s more often than ordinary D.E. Thus most problems in engineering & Science abound with first & Second non linear non homogeneous PDE.

A partial differential eq^n is an eq^n involving two (or more) independent variables x, y & a dependent variable z & its partial derivatives such as $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}$ etc.

i.e, $F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2} \cdots\right) = 0$

Standard Notation

$$p = \frac{\partial z}{\partial x} = z_x$$

$$q = \frac{\partial z}{\partial y} = z_y$$

$$r = \frac{\partial^2 z}{\partial x^2} = z_{xx}$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = z_{xy}$$

$$t = \frac{\partial^2 z}{\partial y^2} = z_{yy}$$

(2) $u = ax\sin y + b\cos x$
[Ans: $u_{xx} + u_{yy} + u = 0$]

(3) $u = ax^3 + by^3$
[Ans: $xp + yq = 3u$]

(4) $a(x^2+y^2) + bu^2 = 1$
[Ans $yp - qp = 0$]

(5) $z = ae^{-b^2 t}\cos bx$
Ans: $z_t = z_{xx}$.

(6) All planes having equal x & y intercepts
$\frac{x}{a} + \frac{y}{a} + \frac{z}{b} = 1$
Ans: $p = q$

ORDER of a PDE is the order of the highest ordered derivative appearing in the P.D.E

If the no of arbitrary constants is equal to no of independent variables, required PDE is of 2 order.

(1) Form the PDE by eliminating a & b from $\log(au-1) = x + ay + b$

$\frac{a}{au-1} \cdot u_x = 1$ & $\frac{a}{au-1} u_y = a$

$\Rightarrow au - 1 = u_y$

$\Rightarrow \frac{a}{u_y} \cdot u_x = 0$

$a = u_y + 1/u$

$\therefore \frac{u_y + 1}{u} \cdot \frac{u_x}{u_y} = 1$

$\Rightarrow u_x(1 + u_y) = u u_y$

FORMATION OF PARTIAL DIFFERENTIAL EQUATIONS

ODE's arise from the elimination of arbitrary constants PDE's can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions from a relation involving three or more variables.

PROBLEMS

1. Form the P.DE by eliminating the arbitrary constant from the eqn $(x-a)^2+(y-b)^2+z^2=4$

Soln:- Consider $(x-a)^2+(y-b)^2+z^2=4$ —— (1)

diff (1) w.r.t x & y partially

$$2(x-a)+2z\frac{\partial z}{\partial x}=0 \Rightarrow (x-a)=-z\frac{\partial z}{\partial x}$$

$$\Rightarrow x-a=-zp \quad \text{——(2)}$$

$$2(y-b)+2z\frac{\partial z}{\partial y}=0 \Rightarrow y-b=-z\frac{\partial z}{\partial y}$$

$$y-b=-zq \quad \text{——(3)}$$

(2) & (3) in (1) we get

$$(-zp)^2+(-zq)^2+z^2=4$$

$z^2(p^2+q^2+1)=4$ which is required PDE

2. $2z=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ —— (1)

Soln:- Diff w.r.t x & y partially

$$\left.\begin{aligned} 2\frac{\partial z}{\partial x}&=\frac{2x}{a^2} \Rightarrow \frac{1}{a^2}=\frac{1}{x}\frac{\partial z}{\partial x}=\frac{p}{x} \\ 2\frac{\partial z}{\partial y}&=\frac{2y}{b^2} \Rightarrow \frac{1}{b^2}=\frac{1}{y}\frac{\partial z}{\partial y}=\frac{q}{y} \end{aligned}\right\} \quad \text{——(2)}$$

② in ①

$$2z = x^2\left(\frac{p}{x}\right) + y^2\left(\frac{q}{y}\right)$$

$$2z = px + qy$$

3. Find the P.D.E of the family of all spheres whose centres lie on the plane $z=0$ & have a constant radius r

Sol^n :- Let the co-ordinates of centre of the sphere on the plane $z=0$, means in the xy plane be $(a, b, 0)$

a & b are arbitrary constants.

let its constant radius be r.

Eq^n of sphere is

$$(x-a)^2 + (y-b)^2 + (z-0)^2 = r^2 \quad \text{——} ①$$

dif^n w.r.t x & y partially we get

$$2(x-a) + 2z\frac{\partial z}{\partial x} = 0$$

$$\Rightarrow x - a = -z\frac{\partial z}{\partial x} = -zp \quad \text{——} ②$$

$$2(y-b) + 2z\frac{\partial z}{\partial y} = 0$$

$$\Rightarrow y - b = -z\frac{\partial z}{\partial y} = -zq \quad \text{——} ③$$

② & ③ in ① we get

$$(-zp)^2 + (-zq)^2 + z^2 = r^2$$

$$z^2p^2 + z^2q^2 + z^2 = r^2$$

$$z^2(p^2 + q^2 + 1) = r^2$$

4. Find the differential equation of the set of all right circular cones whose axes coincides with z-axis.

Sol^n :- The set of all right circular cones whose axes coincide with z-axis, having semi vertical angle α & vertex at $(0,0,c)$ is given by

$$x^2+y^2=(z-c)^2\tan^2\alpha \quad \text{——— (1)}$$

(c, α are arbitrary constants)

dif^n (1) w.r.t x & y partially

$$2x = 2(z-c)\frac{\partial z}{\partial x}\tan^2\alpha \quad \text{——— (2)}$$

& $$2y = 2(z-c)\frac{\partial z}{\partial y}\tan^2\alpha \quad \text{——— (3)}.$$

X'ly eq^n (2) by 'y' & eq^n (3) by x.

$$xy = (z-c)y\frac{\partial z}{\partial x}\tan^2\alpha \quad \& \quad xy = (z-c)x\frac{\partial z}{\partial y}\tan^2\alpha$$

$$\Rightarrow \quad y(z-c)\frac{\partial z}{\partial x}\tan^2\alpha = x(z-c)\frac{\partial z}{\partial y}\tan^2\alpha$$

$$y\frac{\partial z}{\partial x} = x\frac{\partial z}{\partial y}$$

$$\underline{\underline{yp=xq}}$$

5. Form the P.D.E by eliminating a, b, c from $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

Sol^n :- $$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \quad \text{——— (1)}.$$

dif^n partially w.r.t x & y

$$\frac{2x}{a^2}+\frac{2z}{c^2}\frac{\partial z}{\partial x}=0$$

$$\frac{z}{x}\frac{\partial z}{\partial x} = -\frac{c^2}{a^2} \quad \text{——— (2)}$$

$$\frac{2y}{b^2} + \frac{2z}{c^2}\frac{\partial z}{\partial y} = 0$$

$$\frac{z}{y}\frac{\partial z}{\partial y} = \frac{-c^2}{b^2} \quad \text{——— (3)}$$

diff^n ② again w.r.t x partially

$$\frac{z}{x}\frac{\partial^2 z}{\partial x^2} + \frac{1}{x}\frac{\partial z}{\partial x}\cdot\frac{\partial z}{\partial x} - \frac{1}{x^2}\frac{\partial z}{\partial x} = 0$$

x'ly by x^2, we get

$$zx\frac{\partial^2 z}{\partial x^2} + x\left(\frac{\partial z}{\partial x}\right)^2 - z\frac{\partial z}{\partial x} = 0 \quad \text{Required PDE}$$

III'ly diff^n ③ w.r.t y partially we get

$$zy\frac{\partial^2 z}{\partial y^2} + y\left(\frac{\partial z}{\partial y}\right)^2 - z\frac{\partial z}{\partial y} = 0 \quad \text{Required PDE}$$

6. Form the PDE by eliminating arbitrary function from

$$z = \phi(x+ay) + \psi(x-ay) \quad \text{——} ①$$

Soln :-

diff^n ① partially w.r.t x

$$\frac{\partial z}{\partial x} = \phi'(x+ay) + \psi'(x-ay)$$

Again diff^n w.r.t x partially

$$\frac{\partial^2 z}{\partial x^2} = \phi''(x+ay) + \psi''(x-ay)$$

diff^n ① partially w.r.t y

$$\frac{\partial z}{\partial y} = a\,\phi'(x+ay) - a\,\psi'(x-ay)$$

diff^n again w.r.t y partially

$$\frac{\partial^2 z}{\partial y^2} = a^2\phi''(x+ay) + a^2\psi''(x-ay)$$

$$= a^2\left[\phi''(x+ay) + \psi''(x-ay)\right]$$

$$\boxed{\frac{\partial^2 z}{\partial y^2} = a^2\frac{\partial^2 z}{\partial x^2}} \rightarrow \text{Required PDE}$$

7. Form the PDE $z = y f(x) + x \phi(y)$ where f & ϕ are arbitrary functions.

Sol^n:- Given $z = y f(x) + x \phi(y)$

dif^n partially w.r.t x & y

$$\frac{\partial z}{\partial x} = y f'(x) + \phi(y) \quad \text{——— (1)}$$

$$\frac{\partial z}{\partial y} = f(x) + x \phi'(y) \quad \text{——— (2)}$$

dif^n (1) partially w.r.t x

$$\frac{\partial^2 z}{\partial x^2} = y f''(x)$$

dif^n (2) partially w.r.t y

$$\frac{\partial^2 z}{\partial y^2} = x \phi''(y)$$

& $$\frac{\partial^2 z}{\partial x \partial y} = f'(x) + \phi'(y)$$

Consider $$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = xy f'(x) + x \phi(y) + y f(x) + xy \phi'(y)$$

$$= xy[f'(x) + \phi'(y)] + x \phi(y) + y f(x)$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = xy \frac{\partial^2 z}{\partial x \partial y} + z \quad \rightarrow \text{Required PDE}$$

8. Form the PDE given $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$ ——— (1)

Sol^n:- dif^n (1) partially w.r.t x & y

$$\frac{\partial z}{\partial x} = 2 f'\left(\frac{1}{x} + \log y\right)\left(-1/x^2\right) \Rightarrow 2f'\left(\frac{1}{x} + \log y\right) = -x^2 p$$

$$\frac{\partial z}{\partial y} = 2y + 2 f'\left(\frac{1}{x} + \log y\right) \cdot \frac{1}{y}$$

$$q = 2y - x^2 p \cdot \frac{1}{y}$$

$$yq = 2y^2 - x^2 p$$

$$x^2 p + yq = 2y^2 \quad \text{Required PDE}$$

9. Form the PDE by eliminating the arbitrary function from

$$z = y f(y/x) \quad \text{——— (1)}$$

Sol^n:- dif^n (1) partially w.r.t x & y

$$\frac{\partial z}{\partial x} = y f'(y/x)\ (-1/x^2)$$

$$\Rightarrow x^2 p = -y f'(y/x)$$

w.r.t y $\quad \frac{\partial z}{\partial y} = y f'(y/x) \cdot \frac{1}{x} + f(y/x)$

$$q = y\left(\frac{-x^2 p}{y}\right)\frac{1}{x} + \frac{z}{y}$$

$$q = -xp + \frac{z}{y} \Rightarrow qy + xyp = z \quad \text{Required PDE}$$

10. Eliminating the arbitrary function from $z = e^{ax+by} f(ax-by)$

Sol^n:- Given $z = e^{ax+by} f(ax-by)$ ——— (1)

dif^n (1) partially w.r.t x & y

$$\frac{\partial z}{\partial x} = a e^{ax+by} f(ax-by) + a e^{ax+by} f'(ax-by)$$

$$p = a\left[e^{ax+by} f(ax-by) + e^{ax+by} f'(ax-by)\right] \quad \text{——— (2)}$$

$$\frac{\partial z}{\partial y} = b e^{ax+by} f(ax-by) + (-b) e^{ax+by} f'(ax-by)$$

$$q = b\left[e^{ax+by} f(ax-by) - e^{ax+by} f'(ax-by)\right] \quad \text{——— (3)}$$

x'ly (2) by b & (3) by a & adding

$$bp + aq = ab\left[2e^{ax+by} f(ax \pm by)\right]$$

$$bp + aq = 2abz \rightarrow \text{Required PDE}$$

11. Eliminate the arbitrary functions from $v = \frac{1}{r}\left[f(r-at) + F(r+at)\right]$

Soln:- $\frac{\partial v}{\partial r} = \frac{1}{r}\left[f'(r-at) + F'(r+at)\right] - \frac{1}{r^2}\left[f(r-at)+F(r+at)\right]$

$$r^2 \frac{\partial v}{\partial r} = r\left[f'(r-at) + F'(r+at)\right] - \left[f(r-at)+F(r+at)\right]$$

$$\frac{\partial}{\partial r}\left[r^2 \frac{\partial v}{\partial r}\right] = r\left[f''(r-at) + F''(r+at)\right] + \left[f'(r-at) + F'(r+at)\right]$$
$$- \left[f'(r-at) + F'(r+at)\right]$$

$$\Rightarrow \frac{\partial}{\partial r}\left[r^2 \frac{\partial v}{\partial r}\right] = r\left[f''(r-at) + F''(r+at)\right] \quad \text{---------- (1)}$$

III'ly $\frac{\partial v}{\partial t} = \frac{1}{r}\left[-a f'(r-at) + a F'(r+at)\right]$

$$\frac{\partial^2 v}{\partial t^2} = \frac{1}{r}\left[(-a)^2 f''(r-at) + a^2 F''(r+at)\right]$$

$$= \frac{a^2}{r}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r^2 \frac{\partial v}{\partial r}\right)\right] \quad [\text{from (1)}]$$

$$\Rightarrow \frac{\partial^2 v}{\partial t^2} = \frac{a^2}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial v}{\partial r}\right) \rightarrow \text{Required PDE}$$

NOTE :- The PDE corresponding to the function $\phi(u,v)=0$ where u,v are fu's of x,y,z is of the form $Pp+Qq=R$ which is known as Lagrange's eq^n where P,Q,R are functions of x,y & z given by

$$P=\frac{\partial(u,v)}{\partial(y,z)},\quad Q=\frac{\partial(u,v)}{\partial(z,x)},\quad R=\frac{\partial(u,v)}{\partial(x,y)}$$

12. Form the PDE by eliminating the function f, given

$$f(x^2+2yz,\ y^2+2zx)=0 \qquad \text{——— (1)}$$

Soln :- Let $u=x^2+2yz \qquad v=y^2+2zx$

$$P=\frac{\partial(u,v)}{\partial(y,z)}=\begin{vmatrix}\frac{\partial u}{\partial y} & \frac{\partial u}{\partial z}\\ \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z}\end{vmatrix}=\begin{vmatrix}2z & 2y\\ 2y & 2x\end{vmatrix}=4(zx-y^2)$$

$$Q=\frac{\partial(u,v)}{\partial(z,x)}=\begin{vmatrix}\frac{\partial u}{\partial z} & \frac{\partial u}{\partial x}\\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial x}\end{vmatrix}=\begin{vmatrix}2y & 2x\\ 2x & 2z\end{vmatrix}=4(yz-x^2)$$

$$R=\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}=\begin{vmatrix}2x & 2z\\ 2z & 2y\end{vmatrix}=4(xy-z^2)$$

The PDE corresponding to (1) is of the form

$$Pp+Qq=R$$

$$4(zx-y^2)p+4(yz-x^2)q=4(xy-z^2)$$

$$\Rightarrow (zx-y^2)p+(yz-x^2)q=xy-z^2$$

13. Form the PDE by eliminating the arbitrary fⁿ F from the eqⁿ

$F(x+y+z, xy+z^2) = 0$ ——— (1)

Soln:- Let $u = x+y+z$ $\quad v = xy+z^2$

$$P = \frac{\partial(u,v)}{\partial(y,z)} = \begin{vmatrix} \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ x & 2z \end{vmatrix} = 2z-x$$

$$Q = \frac{\partial(u,v)}{\partial(z,x)} = \begin{vmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial x} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2z & y \end{vmatrix} = y-2z$$

$$R = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ y & x \end{vmatrix} = x-y$$

The PDE Corresponding to (1) is of the form

$$Pp + Qq = R$$

$$(2z-x)p + (y-2z)q = x-y$$

14. Form the PDE from the eqⁿ $\phi(x+y+z, x^2+y^2-z^2) = 0$

$u = x+y+z$ $\quad v = x^2+y^2-z^2$

$$P = \frac{\partial(u,v)}{\partial(y,z)} = -2(y+z) \qquad R = \frac{\partial(u,v)}{\partial(x,y)} = 2(y-x)$$

$$Q = \frac{\partial(u,v)}{\partial(z,x)} = 2(x+z)$$

The Required PDE is $-2(y+z)p + 2(x+z)q = 2(y-x)$

$$\Rightarrow (y+z)p - (x+z)q = x-y$$

LINEAR & NONLINEAR PDE

A PDE of I order is called Linear if it is of first degree in p & q & There occur no products among Themselves. otherwise it is called a Non-linear PDE.

Solution of PDE

The General form of I order PDE is

$$f(x, y, z, p, q) = 0 \quad \text{—(1)}$$

x, y – Independent variable

z – dependent "

$p, q \rightarrow \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

Solution of PDE

Complete Soln

↓

involves 2 arbitary Constants a, b

$f(x, y, z, a, b) = 0$

Satisfied The PDE

General Soln

↓

$b = \phi(a)$

$f(x, y, z, a, \phi(a)) = 0$

Soln involving arbitrary fun ϕ

Particular Soln

↓

obtained from complete soln

by giving particular value to a & b

Singular soln

↓

C.S

$f(x, y, z, a, b) = 0$

dif w.r.t a & b

$\frac{\partial f}{\partial a} = 0, \frac{\partial f}{\partial b} = 0$

Relation obtained by eliminating a, b is called Singular soln

Note:- ① $d(xy) = x\,dy + y\,dx$

② $d(\log(x-y)) = \dfrac{dx-dy}{x-y}$

③ $d(\log(x^2+y^2+z^2)) = \dfrac{2x\,dx+2y\,dy+2z\,dz}{x^2+y^2+z^2}$

④ $d[(x+y+z)^2] = 2(x+y+z)(dx+dy+dz)$.

⑤ Property in ratio & proportion

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \frac{k_1a_1 + k_2a_2 + k_3a_3}{k_1b_1 + k_2b_2 + k_3b_3}$$

⑥ For $\dfrac{dx}{P} = \dfrac{dy}{Q} = \dfrac{dz}{R}$ $\quad l', m', n'$

choose multipliers l, m, n (non-zero) such that

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lp + mQ + nR} = \frac{l'dx + m'dy + n'dz}{l'p + m'Q + n'R}$$

If $Dx = 0$ integrate $Nx\,(l\,dx + m\,dy + n\,dz)$ the resulting eqn is one soln.

Solution Of The Equation Of The Type $Pp + Qq = R$

(Langrage's Equation)

Working rule for finding the sol^n of Lagrange's Linear eq^n $Pp + Qq = R$

Step 1 : Form the auxillary eq^n (AE) $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Step 2 : Consider the pair or Combination such that it is integrable

$$\frac{dx}{P} = \frac{dy}{Q} \quad \text{or} \quad \frac{dy}{Q} = \frac{dz}{R} \quad \text{or} \quad \frac{dz}{R} = \frac{dx}{P}$$

[Suppose $\frac{dx}{P} = \frac{dy}{Q}$ is integrable then we get the relation b/w x & y which is a part of the general sol^n.]

Step 3 : General or Complete sol^n is given by

$$\phi(u, v) = 0 \quad \text{or} \quad \phi(c_1, c_2) = 0 .$$

Problems

(1) Solve $x^2 p + y^2 q = z^2$

Sol^n :- Given eq^n is of the form $Pp + Qq = R$

$$P = x^2 , \quad Q = y^2 , \quad R = z^2$$

A.E are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$$

Consider, $\frac{dx}{x^2} = \frac{dy}{y^2}$, on integration, $\frac{-1}{x} = \frac{-1}{y} + c_1$

$$\text{or} \quad c_1 = \frac{1}{y} - \frac{1}{x}$$

Consider, $\frac{dy}{y^2} = \frac{dz}{z^2}$ on Integration, $\frac{-1}{y} = \frac{-1}{z} + c_2$

$$\Rightarrow c_2 = \frac{1}{z} - \frac{1}{y}$$

$\therefore$ General sol^n is $\phi\left(\frac{1}{y} - \frac{1}{x}, \frac{1}{z} - \frac{1}{y}\right) = 0$

2. Solve $p\cot x + q\cot y = \cot z$

Sol^n: Given eq^n is of the form $Pp+Qq=R$

$P = \cot x$, $Q = \cot y$, $R = \cot z$

A.E, $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{\cot x} = \frac{dy}{\cot y} = \frac{dz}{\cot z}$$

$$\tan x\,dx = \tan y\,dy = \tan z\,dz$$

Consider $\tan x\,dx = \tan y\,dy$

Integrating, $\log \sec x = \log \sec y + c_1$

$$\Rightarrow \log\left(\frac{\sec x}{\sec y}\right) = c_1$$

$$\Rightarrow \sec x \cos y = e^c = c_1$$

III^{ly} $\tan y\,dy = \tan z\,dz$

Integ, $\Rightarrow \log \sec y = \log \sec z + c'$

$$\Rightarrow \log\left(\frac{\sec y}{\sec z}\right) = c'$$

$$\Rightarrow \sec y \cos z = e^{c'} = c_2$$

$\therefore$ General Sol^n is $\phi(c_1, c_2) = 0$

$$\phi(\sec x \cos y, \sec y \cos z) = 0$$

3. Solve $(y^2+z^2)p + x(yq - z) = 0$

Sol^n: Given $(y^2+z^2)p + xyq - xz = 0$

$$(y^2+z^2)p + xyq = xz$$

$$\Rightarrow Pp + Qq = R$$

A.E are, $\frac{dx}{y^2+z^2} = \frac{dy}{xy} = \frac{dz}{xz} = k$ ——— (1)

Consider, $\frac{dy}{xy} = \frac{dz}{xz}$ or $\frac{dy}{y} = \frac{dz}{z}$

$$\log y = \log z + c$$

$$\log\left(\frac{y}{z}\right) = c$$

$$\frac{y}{z} = e^c \Rightarrow \boxed{\frac{y}{z} = c_1}$$

Using Multipliers, $x, -y, -z$ in (1) we get

$$\frac{x\,dx}{x(y^2+z^2)} = \frac{-y\,dy}{-y(xy)} = \frac{-z\,dz}{-z(xz)} = k$$

$$x\,dx - y\,dy - z\,dz = k\left[x(y^2+z^2) - y(xy) - z(xz)\right]$$

$$x\,dx - y\,dy - z\,dz = 0$$

Integrating, $\frac{x^2}{2} - \frac{y^2}{2} - \frac{z^2}{2} = c'$

$$x^2 - y^2 - z^2 = 2c' = c_2$$

∴ General sol^n is $\phi\left(\frac{y}{z}, x^2-y^2-z^2\right) = 0$

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A. 4. Solve $x(y^2-z^2)p + y(z^2-x^2)q = z(x^2-y^2)$

ar Sol^n:- Given eq^n is of the form $Pp + Qq = R$

A.E are $\frac{dx}{x(y^2-z^2)} = \frac{dy}{y(z^2-x^2)} = \frac{dz}{z(x^2-y^2)} = k$ ——— (1)

Consider, $\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = k\left[(y^2-z^2) + (z^2-x^2) + (x^2-y^2)\right]$

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

On integration, $\log x + \log y + \log z = c$

$$\Rightarrow \log(xyz) = c$$

$$xyz = e^c = c_1$$

Consider, $x\,dx + y\,dy + z\,dz = k\left[x^2(y^2-z^2) + y^2(z^2-x^2) + z^2(x^2-y^2)\right]$

$$x\,dx + y\,dy + z\,dz = 0$$

on integration, $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c'$

$$x^2 + y^2 + z^2 = 2c' = c_2$$

General soln is $\phi(xyz, x^2+y^2+z^2) = 0$

5. Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Soln :- Given eqn is of the form $Pp + Qq = R$

A.E are $\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = k$ (say)

Consider $\frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2} = k\left[(y-z) + (z-x) + (x-y)\right]$

$$x^{-2}dx + y^{-2}dy + z^{-2}dz = 0$$

On integration, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -c = c_1$

Consider $\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = k\left[x(y-z) + y(z-x) + z(x-y)\right]$

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

on integration, $\log x + \log y + \log z = c'$

$$\log(xyz) = c'$$

$$xyz = c_2$$

$\therefore$ General soln is $\phi\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0$

6. Solve $y^2zp = x^2(zq+y)$

Soln :- Given eqn is of the form $Pp + Qq = R$

A.E are $$\frac{dx}{y^2z} = \frac{dy}{-x^2z} = \frac{dz}{x^2y}$$

Consider $$\frac{dx}{y^2z} = \frac{dy}{-x^2z}$$

$$\Rightarrow x^2dx + y^2dy = 0$$

on integration, $$\frac{x^3}{3} + \frac{y^3}{3} = c$$

$$\Rightarrow x^3 + y^3 = 3c = c_1$$

III^ly Consider $$\frac{dy}{-x^2z} = \frac{dz}{x^2y}$$

$$ydy + zdz = 0$$

on integration, $$\frac{y^2}{2} + \frac{z^2}{2} = c$$

$$\Rightarrow y^2 + z^2 = 2c = c_2$$

General soln is $\phi(x^3+y^3, y^2+z^2) = 0$

7. Solve $(x^2-yz)p + (y^2-zx)q = z^2-xy$

soln :- Given eqn is of the form $Pp+Qq=R$

A.E are, $$\frac{dx}{x^2-yz} = \frac{dy}{y^2-zx} = \frac{dz}{z^2-xy} = k \text{ (say)}$$

Consider $dx - dy = k[(x^2-yz) - (y^2-zx)]$

$$= k[(x-y)(x+y+z)]$$

$$\frac{dx-dy}{(x-y)(x+y+z)} = k$$

Wkt $\frac{dy-dz}{(y-z)(x+y+z)} = k$, $\frac{dz-dx}{(z-x)(x+y+z)} = k$

$$\frac{dx-dy}{(x-y)(x+y+z)} = \frac{dy-dz}{(y-z)(x+y+z)} = \frac{dz-dx}{(z-x)(x+y+z)}$$

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$$

Consider $\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$

Integrating, $\log(x-y) = \log(y-z) + c$

$$\log\left[\frac{x-y}{y-z}\right] = c$$

$$\frac{x-y}{y-z} = e^c = c_1$$

Consider $\frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$

Integrating, $\log(y-z) = \log(z-x) + c$

$$\log\left[\frac{y-z}{z-x}\right] = c$$

$$\Rightarrow \frac{y-z}{z-x} = e^c = c_2$$

$\therefore$ General Soln is $\phi\left[\frac{x-y}{y-z}, \frac{y-z}{z-x}\right] = 0$

8. Solve $(y+z)p+(z+x)q = x+y$

Soln:- Given eqn is of the form $Pp+Qq=R$

A.E's are $\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} = k$

Consider $dx-dy = k[(y+z)-(z+x)] = (-k)(x-y)$

$$\frac{dx-dy}{x-y} = -k$$

IIIly $\frac{dy-dz}{y-z} = -k$ & $\frac{dz-dx}{z-x} = -k$

$$\Rightarrow \frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$$

Consider $\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$

Integrating, $\log(x-y) = \log(y-z) + c$

$$\log\left[\frac{x-y}{y-z}\right] = c$$

$$\frac{x-y}{y-z} = e^c = c_1$$

Consider $\frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$

Integrating, $\log(y-z) = \log(z-x) + c$

$$\log\left[\frac{y-z}{z-x}\right] = c$$

$$\frac{y-z}{z-x} = e^c = c_2$$

$\therefore$ General soln is $\phi\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0$

(1) $(y-x)p+(y+x)q=(x+y)/u$

$\Rightarrow \frac{dx+dy}{2y}=\frac{dy}{y+x}$ $\Rightarrow (x+y)(dx+dy)=2y\,dy$.

$\frac{(x+y)^2}{2}=y^2+c_1$

$x\,dx-y\,dy+u\,du=0 \Rightarrow x^2-y^2+u^2=c_2$.

(2) $(mu-ny)u_x+(nx-lu)u_y=ly-mx$

$x\,dx+y\,dy+u\,du=0$

$l\,dx+m\,dy+n\,du=0$

Ans: $F(x^2-y^2,\ x-y+u)=0$

(3) $x\,u_x+y\,u_y=x-y$.

(4) ~~$(y-z)u_x+(z-x)u_y+(x-y)u_z=0$~~ ~~Ans: F(~~

Find the first integrals of the following symmetric systems of D.E

1) $\frac{dx}{x(y^2-u^2)}=\frac{dy}{-y(x^2+u^2)}=\frac{du}{u(x^2+y^2)}$

Ans: $x^2+y^2+u^2=c_1$

$\frac{yu}{x}=c_2$

2) $\frac{dx}{-x^2}=\frac{dy}{xy-2u^2}=\frac{du}{xu}$

$xu=c_1$

$xy+u^2=c_2$

(5) $(x^2+y^2)u_x+2xy\,u_y=-u^2$

Ans: $\frac{dx+dy}{(x+y)^2}=\frac{-du}{u^2}$

$\frac{dx-dy}{(x-y)^2}=\frac{-du}{u^2}$

$F\left(\frac{1}{x+y}+\frac{1}{u},\ \frac{1}{x-y}+\frac{1}{u}\right)=0$

$px(x+y)=qy(x+y)-(2x+2y+z)(x-y)$

$\frac{dx}{x}=\frac{dy}{y} \Rightarrow xy=c_1$

$\frac{dx+dy}{(x+y)(x-y)}=\frac{dz}{-(x-y)(2x+2y+z)} \Rightarrow$

$(x+y+z)\,d(x+y)+(x+y)\,d(x+y+z)=0$

Solution Of Partial Differential Equations By Method Of Separation Of Variables

Several problems in fluid mechanics, solid mechanics, heat transfer, electromagnetic theory & other areas of physics are modeled as Initial Boundary Value Problems (IBVP) consisting of partial D.E & initial conditions or Boundary conditions. Method of separation of variables is a powerful tool to solve such IBVP when PDE is linear & boundary conditions are homogeneous.

Separation of variables is a powerful technique to solve PDE. For a PDE, in the fun z of two independent variables x & y, assume that the required soln is separable i.e,

$$z(x,y) = X(x)\,Y(y) \qquad \text{——— (1)}$$

where $X(x)$ is a fun of x alone

$Y(y)$ is a fun of y alone.

Then substitution of z from (1) & its derivatives reduces the PDE to two ordinary D.E, which are solved & their product is considered. This method was derived by D'Alembert.

The determination of soln of PDE reduces to the determination of soln's to two D.DE with appropriate Conditions.

PROBLEMS

1. Use the method of separation of variables to solve

$$\frac{\partial z}{\partial x} = 2\frac{\partial z}{\partial y} + z \quad \text{where} \quad z(x,0) = 6e^{-3x}$$

Sol^n :- Given $\frac{\partial z}{\partial x} = 2\frac{\partial z}{\partial y} + z$ ——— (1)

Let Sol^n of (1) be

$$z = X(x)\,Y(y) = XY \quad \text{——— (2)}$$

$$\therefore \frac{\partial z}{\partial x} = X'Y \;,\; \frac{\partial z}{\partial y} = XY'$$

$\therefore$ (1) becomes

$$X'Y = 2XY' + XY$$

Separating the variables, we get

$$\frac{X'-X}{2X} = \frac{Y'}{Y} = k$$

$$\Rightarrow \frac{X'-X}{2X} = k \quad \& \quad \frac{Y'}{Y} = k$$

$$\Rightarrow X'-X-2Xk=0 \quad \& \quad Y'-Yk=0$$

Consider $X'-X-2Xk=0$

$$\Rightarrow (D-1-2k)X = 0$$

A.E $\Rightarrow m-1-2k=0$

$$\Rightarrow m = 1+2k$$

$$\therefore X = c_1 e^{(1+2k)x}$$

Consider $Y'-Yk=0$

$$(D-k)Y=0$$

A.E $\Rightarrow m-k=0$

$$m=k$$

$$\therefore Y = c_2 e^{ky}$$

$\therefore$ (2) becomes

$$z = XY = c_1c_2\, e^{(1+2k)x}\, e^{ky}$$

Given $y=0$, $z = 6e^{-3x}$

$$\Rightarrow 6e^{-3x} = c_1c_2\, e^{(1+2k)x}$$

$$\Rightarrow c_1c_2 = 6 \quad \& \quad 1+2k = -3 \Rightarrow k = -2$$

$\therefore \quad z = 6\, e^{-3x}\, e^{-2y}$

(2) Solve $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$

Sol^n :- Given $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$ ——— (1)

Let $u = XY$ be the sol^n of (1)

$\therefore \frac{\partial u}{\partial x} = X'Y \quad , \quad \frac{\partial u}{\partial y} = XY'$

(1) becomes $x^2 X'Y + y^2 XY' = 0$

Separating the variables we get

$$\frac{x^2}{X}\frac{dX}{dx} = \frac{-y^2}{Y}\frac{dY}{dy} = k$$

$$\frac{x^2}{X}\frac{dX}{dx} = k \quad , \quad \frac{-y^2}{Y}\frac{dY}{dy} = k$$

$$\frac{dX}{X} = \frac{k}{x^2}\,dx \quad , \quad \frac{dY}{Y} = \frac{-k}{y^2}\,dy$$

Integrating

$\log X = -\frac{k}{x} + c_1$ $\qquad$ $\log Y = \frac{k}{y} + c_2$

$\Rightarrow X = e^{-\frac{k}{x} + c_1}$ $\qquad$ $Y = e^{\frac{k}{y} + c_2}$

$\therefore \quad u = XY$

$= e^{c_1 + c_2}\, e^{-\frac{k}{x} + \frac{k}{y}}$

$= C\, e^{-\frac{k}{x} + \frac{k}{y}}$ $\qquad$ where $e^{c_1 + c_2} = C$

3. Solve $\frac{\partial^2 z}{\partial x^2} = a^2 z$ given $x=0$, $\frac{\partial z}{\partial x} = a\sin y$ & $\frac{\partial z}{\partial y} = 0$

Soln :- Let the soln of the given eqn be

$$z = XY \quad \text{——(1)}$$

Given eqn becomes

$$X''Y = a^2 XY$$

$$\frac{X''}{X} = a^2$$

$$X'' - a^2 X = 0$$

A.E $\Rightarrow m^2 - a^2 = 0 \Rightarrow m = \pm a$

$$\therefore X = c_1 e^{ax} + c_2 e^{-ax}$$

$\because z$ is a fn of x & y

Soln is of the form

$$z = c_1(y) e^{ax} + c_2(y) e^{-ax} \quad \text{——(2)}$$

$$\frac{\partial z}{\partial x} = c_1(y)\, a\, e^{ax} + c_2(y)(-a) e^{-ax}$$

when $x=0$, $\frac{\partial z}{\partial x} = a\, c_1(y) - a\, c_2(y)$

$$\Rightarrow a[c_1(y) - c_2(y)] = a\sin y$$

$$c_1(y) - c_2(y) = \sin y \quad \text{——(3)}$$

Wkly $\frac{\partial z}{\partial y} = c_1'(y) e^{ax} + c_2'(y) e^{-ax}$

$x=0$, $\frac{\partial z}{\partial y} = c_1'(y) + c_2'(y)$

$$\Rightarrow c_1'(y) + c_2'(y) = 0$$

Integrating, $c_1(y) + c_2(y) = k \quad \text{——(4)}$

Solving (3) & (4) $c_1 = \frac{1}{2}[k + \sin y]$, $c_2 = \frac{1}{2}[k - \sin y]$

$$\therefore z = \frac{1}{2}(k + \sin y) e^{ax} + \frac{1}{2}(k - \sin y) e^{-ax}$$

4) Solve $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ $\left[X = c_1 e^{(1+\sqrt{1+k})x} + c_2 e^{(1-\sqrt{1+k})x}, \; Y = c_3 e^{-ky} \right]$

5) Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$, $x=0$, $\frac{\partial z}{\partial x} = 1$ $\left[z = e^y \cos x + \sin x \right]$

6) Solve $\frac{\partial^2 u}{\partial x^2} - 4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ $\left[u = \left(A e^{(2+\sqrt{4-k})x} + B e^{(2-\sqrt{4-k})x} \right) e^{-ky} \right]$

7) Solve $4u_x + u_y = 3u$, $u(0,y) = e^{-5y}$ $\left[u = e^{2x-5y} \right]$

8) Solve $3u_x + 2u_y = 0$, $u(x,0) = 4e^{-x}$ $\left[u = 4e^{-\frac{1}{2}(2x-3y)} \right]$

9) Solve $u_{xt} = e^{-t}\cos x$, $u(x,0) = 0$ $\left[u = \sin x - e^{-t}\sin x \right]$

$\frac{\partial u}{\partial t}(x,0) = e^{-t}$.

$x'T' = e^{-t}\cos x$.

10) Solve $u_{xx} = u_y + 2u$, $u(0,y) = 0$ $\left[u = \frac{1}{\sqrt{k}} \sinh\sqrt{k}x + e^{-3y}\sin x \right] \sinh\sqrt{k}$

$\frac{\partial u}{\partial x}(0,y) = 1 + e^{-3y}$

11) Solve $u_x = 2u_t + u$, $u(x,0) = 6e^{-3x}$ $\left[u = 6e^{-(3x+2t)} \right]$.

12) Solve $z_{xx} - 2z_x + z_y = 0$ $\left\{ u = \left[c_1 e^{(1+\sqrt{1+k})x} + c_2 e^{(1-\sqrt{1+k})x} \right] e^{-ky} \right.$

13) Solve $u_{xx} - u_y = 0$

$u(x,0) = 4e^{-3x} - 6e^{-5x}$

14) $4u_x - u_y = 0$

$4X'Y - XY' = 0$.

$4\frac{X'}{X} = \frac{Y'}{Y} = k$.

$u = C e^{\frac{k}{4}x} \cdot e^{ky}$.

$4e^{-3x} - 6e^{-5x} = C e^{\frac{k}{4}x}$.

$4X' = kX$

$X(x) = c_1 e^{k/4 x}$ $Y(y) = c_2 e^{ky}$.

$C = \frac{4e^{-3x} - 6e^{-5x}}{e^{\frac{k}{4}x}}$.

$$u = \left(4e^{-3x} - 6e^{-5x}\right) e^{ky}.$$

$$X''y = XY' + 2XY.$$

$$\frac{X''}{X} = \frac{Y'}{Y} + 2 = k.$$

$$\frac{X''}{X} = k \qquad \text{let } k = p^2$$

$$X'' - p^2 X = 0.$$

$$m^2 = p^2$$

$$m = \pm p.$$

$$\frac{Y'}{Y} + (2-k) = 0.$$

$$m = -(2-k)$$

$$\therefore X(x) = \left(C_1 e^{px} + C_2 e^{-px}\right) C_3 e^{(p^2-2)y} \Rightarrow \left(A e^{px} + B e^{-px}\right) e^{(p^2-2)y}$$

when $x = 0$.

$$0 = (C_1 + C_2) C_3 \; (A+B) \; e^{(p^2-2)y} \Rightarrow \boxed{A+B=0}$$

$$\frac{\partial u}{\partial x} = p\left[A e^{px} - B e^{-px}\right] e^{(p^2-2)y}.$$

$$1 + e^{-3y} = p\left[A - B\right] e^{(p^2-2)y}.$$

$$A - B = \frac{1 + e^{-3y}}{p\, e^{(p^2-2)y}}.$$

$$A = \frac{1 + e^{-3y}}{2p\, e^{(p^2-2)y}}.$$

$$B = -\frac{1}{2}\left[\frac{1 + e^{-3y}}{p\, e^{(p^2-2)y}}\right]$$

SOLUTION OF THE WAVE EQUATION BY THE METHOD OF SEPARATION OF VARIABLES

*

Let the wave eqⁿ be $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ ——— (1)

Let the soⁿ of (1) be

$$y = X(t)T(t)$$

$$\frac{\partial^2 y}{\partial t^2} = XT'' \quad , \quad \frac{\partial^2 y}{\partial x^2} = X''T$$

(1) becomes $XT'' = c^2 X''T$

$$\frac{X''}{X} = \frac{T''}{c^2 T} = k$$

$\Rightarrow X'' - Xk = 0$ & $T'' - c^2 Tk = 0$

$\frac{d^2X}{dx^2} - Xk = 0$ & $\frac{d^2T}{dt^2} - Tc^2k = 0$

Consider $\frac{d^2X}{dx^2} - Xk = 0$

A.E $\Rightarrow m^2 - k = 0$ ——— (1)

(i) When k is positive, let $k = p^2$

(1) becomes $m^2 - p^2 = 0 \Rightarrow m = \pm p$

$$\therefore X = C_1 e^{px} + C_2 e^{-px}$$

(ii) When k is negative, let $k = -p^2$

(1) $\Rightarrow m^2 + p^2 = 0$

$m = \pm pi$

$$\therefore X = C_3 \cos px + C_4 \sin px$$

(iii) When $k=0$

$① \Rightarrow m^2=0 \Rightarrow m=0,0$

$\therefore X=(C_5+C_6x)e^{0x}$

Now Consider $\frac{d^2T}{dt^2}-Tc^2k=0$

A.E $\Rightarrow m^2-c^2k=0$ ——— (2)

(i) When k is positive, let $k=p^2$

$② \Rightarrow m^2=c^2p^2$

$m=\pm cp$

$\therefore T=C_7e^{cpt}+C_8e^{-cpt}$

(ii) When k is negative, let $k=-p^2$

$② \Rightarrow m^2=\pm cpi$

$\therefore T=C_9\cos cpt+C_{10}\sin cpt$

(iii) When $k=0$

$② \Rightarrow m=0,0$

$\therefore T=(C_{11}+C_{12}t)e^{0t}$

Thus Various possible soln of the wave eqn are

$y=(C_1e^{px}+C_2e^{-px})(C_7e^{cpt}+C_8e^{-cpt})$

$y=(C_3\cos px+C_4\sin px)(C_9\cos cpt+C_{10}\sin cpt)$

$y=(C_5+C_6x)(C_{11}+C_{12}t)$

ONE DIMENSIONAL HEAT EQUATION

One dimensional heat-flow equation is given by

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{——— (1)}$$

where c^2 is called the diffusivity of the substance.

Assume the sol^n of (1) is of the form

$$u(x,t) = X(x)\,T(t)$$

where X is a fu^n of x alone
T is a fu^n of t only.

Substituting in (1)

$$XT' = c^2 X''T$$

$$\frac{X''}{X} = \frac{T'}{c^2 T} = k$$

$$\Rightarrow \frac{X''}{X} = k \quad \& \quad \frac{T'}{c^2 T} = k$$

$$X'' - kX = 0 \quad \& \quad T' - kc^2 T = 0 \quad \text{——— (2)}$$

Solving.

Case(i):- When k is positive, $= p^2$

A.E $\Rightarrow m^2 - p^2 = 0$

$m = \pm p$

$\therefore X(x) = C_1 e^{px} + C_2 e^{-px}$

A.E $\Rightarrow m - c^2 p^2 = 0$

$m = c^2 p^2$

$\therefore T(t) = C_3 e^{c^2 p^2 t}$

Case(ii) :- When k is negative, $= -p^2$ (say)

A.E $\Rightarrow m^2 + p^2 = 0$

$m^2 = -p^2$

$m = \pm pi$

$X(x) = C_4 \cos px + C_5 \sin px$

A.E $\Rightarrow m + p^2c^2 = 0$.

$m = -p^2c^2$

$\therefore T(t) = C_6 e^{-p^2c^2t}$.

Case(iii) :- When k is zero

A.E $\Rightarrow m^2 = 0$.

$m = 0, 0$

A.E $\Rightarrow m = 0$.

$T(t) = C_9$

$X(x) = C_7 + C_8 x$.

Thus the various possible solutions of the heat eq^n are

$$u = \left(C_1 e^{px} + C_2 e^{-px}\right) C_3 e^{c^2p^2t}$$

$$u = \left(C_4 \cos px + C_5 \sin px\right) C_6 e^{-c^2p^2t}$$

$$u = \left(C_7 + C_8 x\right) C_9$$

SOLUTION OF TWO DIMENSIONAL LAPLACE'S EQUATION

Laplace eq^n is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ ——— (1)

Let the sol^n of (1) be $u = XY$

$$\frac{\partial^2 u}{\partial x^2} = X''Y \qquad \frac{\partial^2 u}{\partial y^2} = XY''$$

$\therefore$ (1) becomes $X''Y + XY'' = 0$

$$\Rightarrow \frac{X''}{X} = \frac{-Y''}{Y} = k \text{ (say)}$$

$$X'' - Xk = 0 \quad \& \quad Y'' + Yk = 0$$

or $\frac{d^2X}{dx^2} - Xk = 0 \quad \& \quad \frac{d^2Y}{dy^2} + Yk = 0$

Consider $\frac{d^2X}{dx^2} - Xk = 0$

A.E $\Rightarrow m^2 - k = 0$ ——— (2)

Case (i) :- when k is positive, let $k = p^2$

(2) becomes $m^2 - p^2 = 0 \Rightarrow m = \pm p$

$$\therefore X = C_1 e^{px} + C_2 e^{-px}$$

Case (ii) :- When k is negative let $k = -p^2$

(2) becomes $m^2 + p^2 = 0 \Rightarrow m = \pm ip$

$$\therefore X = C_3 \cos px + C_4 \sin px$$

Case (iii) :- when $k = 0$

(2) becomes $m^2 = 0 \quad \therefore m = 0, 0$

$$\therefore X = (C_5 + C_6 x) e^{0x}$$

Consider $\frac{d^2Y}{dy^2} + Yk = 0$

A.E $\Rightarrow m^2 + k = 0$ ——— (3)

Case (i) :- when k is positive let $k = p^2$

(3) becomes $m^2 + p^2 = 0 \Rightarrow m = \pm pi$

$\therefore Y = C_7 \cos py + C_8 \sin py$

Case (ii) :- When k is negative, let $k = -p^2$

(3) becomes $m^2 - p^2 = 0 \Rightarrow m = \pm p$

$\therefore Y = C_9 e^{py} + C_{10} e^{-py}$

Case (iii) :- when $k = 0$

(3) becomes $m^2 = 0 \Rightarrow m = 0, 0$

$Y = (C_{11} + C_{12} y) e^{0y}$

The various possible soln's of (1) are

$u = (C_1 e^{px} + C_2 e^{-px})(C_7 \cos py + C_8 \sin py)$

$u = (C_3 \cos px + C_4 \sin px)(C_9 e^{py} + C_{10} e^{-py})$

$u = (C_5 + C_6 x)(C_{11} + C_{12} y)$

PARTIAL DIFFERENTIAL EQUATIONS

CLASSIFICATION OF SECOND ORDER PDE

The classification of PDE is motivated by the classification of the quadratic equation of the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

which is elliptic, parabolic or hyperbolic according as the discriminant $B^2 - 4AC$ is negative, zero or positive.

Thus, we have the following second order linear PDE in two variables x & y.

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$$

where the coefficients A, B, C, D may be function of x & y.

$B^2 - 4AC < 0 \rightarrow$ Elliptic

$B^2 - 4AC = 0 \rightarrow$ Parabolic

$B^2 - 4AC > 0 \rightarrow$ Hyperbolic

Ex:-

(1) Laplace equation in two dimension $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

$A = 1, \quad B = 0, \quad C = 1 \quad ; \quad B^2 - 4AC = 0 - 4 = -4 < 0$

Equation is Elliptic.

(2) One Dimensional Wave equation $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$

$A = a^2, \quad B = 0, \quad C = -1 \quad ; \quad B^2 - 4AC = 0 + 4a^2 > 0$

Given PDE is Hyperbolic.

(3) One Dimensional Heat equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$

$A = a^2, \quad B = 0, \quad C = 0 \quad ; \quad B^2 - 4AC = 0 - 0 = 0$

Give PDE is Parabolic

(4) $3u_{xx} + 10u_{xy} + 3u_{yy} = 0$

$A=3,\ B=10,\ C=3\ ;\ B^2-4AC = 100-36 = 64 > 0$

Given PDE is Hyperbolic

(5) $x^2 u_{xx} - 2xy\, u_{xy} + y^2 u_{yy} = e^x$

$A=x^2,\ B=-2xy,\ C=y^2\ ;\ B^2-4AC = 4x^2y^2 - 4x^2y^2 = 0$

Given PDE is Parabolic.

(6) $u_{xx} + x^2 u_{yy} = 0$

$A=1,\ B=0,\ C=x^2\ ;\ B^2-4AC = -4x^2 < 0$

Given PDE is Elliptic

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(7) $y^2 u_{xx} - 2xy\, u_{xy} + x^2 u_{yy} = \frac{y^2}{x} u_x + \frac{x^2}{y} u_y$ → Parabolic

(8) $(1+x^2) u_{xx} + (1+y^2) u_{yy} + x u_x + y u_y = 0$ → Elliptic

(9) $u_{xx} - 2\sin x\, u_{xy} - \cos^2 x\, u_{yy} + \cos x\, u_y = 0$ → Hyperbolic

(10) $u_{xx} + x u_{yy} = 0,\ x \neq 0 \Rightarrow B^2-4AC = -4x$

$x<0$ → Hyperbolic

$x>0$ → Elliptic

(11) $u_{xx} + 2u_{xy} + \sin^2 x\, u_{yy} + u_y = 0$

$B^2-4AC = 4\cos^2 x,\ \forall x \neq \frac{(2n-1)\pi}{2}$ given PDE is Hyperbolic

(12) $y u_{xx} + (x+y) u_{xy} + x u_{yy} = 0$

$B^2-4AC = (x+y)^2 - 4xy = (x-y)^2 > 0$

Given PDE is Hyperbolic everywhere except along the line $y=x$

$y=x$, it is Parabolic.

(13) $u_{xx} + \frac{2N}{x} u_x = \frac{1}{a^2} u_{tt}$, N & a are Constants

→ Hyperbolic

Various possible solutions of 1-D Heat Eqn

1. Using Variable seperable method :

Consider the 1-D Heat Eqn

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \longrightarrow \text{III}$$

Let $u(x,t) = XT$ — 2 be the soln of III where X & T are functions of x & t only

$$\frac{\partial u}{\partial t} = XT' \quad \& \quad \frac{\partial^2 u}{\partial x^2} = X''T$$

Substituting in III, we get

$$XT' = c^2 X''T$$

Seperating the variables we get

$$\frac{X''}{X} = \frac{1}{c^2}\frac{T'}{T}$$

Equating both sides to constant K,

$$\frac{X''}{X} = K \quad \text{and} \quad \frac{1}{c^2}\frac{T'}{T} = K$$

$$\Rightarrow X'' = KX \quad \& \quad T' = c^2 KT$$

$$\Rightarrow (X'' - KX) = 0 \quad \& \quad T' - c^2 KT = 0$$

$$\Rightarrow (D^2 - K)X = 0 \longrightarrow 3 \quad \& \quad (D - c^2 K)T = 0 \longrightarrow 4$$

where $D = \frac{d}{dx}$ $\qquad D = \frac{d}{dt}$

→ Hyperbolic

Case (i) when $K = 0$.

From [3], $D^2X = 0$

The A.E is given by

$m^2 = 0$

$\Rightarrow m = 0, 0.$

$\therefore X = (C_1 + C_2 x) e^{0x}$

$DT = 0$

The A.E.

$\Rightarrow m = 0$

$\therefore T = C_3 e^{0t}$

substituting in [2] we get

$$u(x,t) = (C_1 + C_2 x) C_3 = A + Bx,$$ is the

Required soln where $A = C_1 C_3$, $B = C_2 C_3$

Case (ii): when K is positive, $K = +p^2$

From [3]

$(D^2 - p^2) X = 0$

The A.E. is given by

$m^2 - p^2 = 0$

$m^2 = p^2$

$m = \pm\sqrt{p^2} = \pm p$

$\therefore X = C_4 e^{px} + C_5 e^{-px}$

$(D - c^2p^2) T = 0$

$m - c^2p^2 = 0$

$\Rightarrow m = c^2p^2$

$T = C_6 e^{c^2p^2 t}$,

substituting in [2], we get

$u(x,t) = (C_4 e^{px} + C_5 e^{-px}) C_6 e^{c^2p^2 t}$

(or)

$$u(x,t) = (C e^{px} + D e^{-px}) e^{c^2p^2t}$$ is the required soln, where $C = c_4c_6$ & $D = c_5c_6$

Case(iii): when K is -ve, $K = -p^2$

From [3]

$$(D^2 + p^2)X = 0 \qquad (D + c^2p^2)T = 0$$

The A.E is given by

$$m + c^2p^2 = 0$$

$$m^2 + p^2 = 0$$

$$\Rightarrow m^2 = -p^2 \Rightarrow m = -c^2p^2$$

$$\Rightarrow m = \pm\sqrt{-p^2} = \pm ip = \alpha \pm i\beta$$

$$\therefore X = e^{0x}[c_7 \cos\beta x + c_8 \sin\beta x], \quad T = c_9 e^{-c^2p^2t}$$

Substituting in [1] we get

$$u(x,t) = (c_7 \cos\beta x + c_8 \sin\beta x)\, c_9\, e^{-c^2p^2t}$$

(or)

$$u(x,t) = (E\cos\beta x + F\sin\beta x)\, e^{-c^2p^2t}$$ is the required soln, where $E = c_7c_9$ & $F = c_8c_9$

Of these three Solns, we have to choose that solution which is consistent with the physical nature of the problem.

As we are dealing with problems on Heat conduction, It must be a periodic function of x & t.

Hence this soln must Involve trigonometric terms. Accordingly the solution is

$$u(x,t) = (A\cos\beta x + B\sin\beta x)\, e^{-c^2\beta^2 t}$$

is the only suitable soln.